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Physico-chemical analysis of water of Kanke dam, Ranchi, Jharkhand.

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Abstract : The present work is an attempt to analyse various physicochemical factors of the water of Kanke Dam and its suitability for human consumption and public hygiene scenario and health impacts. Safe drinking water is the primary need of time, as most of the people depends upon groundwater sources that have problems of hardness, calcium, Nitrate, Phosphate, fluoride, D.O., B.O.D., COD. Different parameters of water of Kanke Dam has been analyzed and evaluated, following the standard methods recommended by APHA-AWWA-WPCF (1981).[1] In the present investigation some physico-chemical parameters are observed within the prescribed limits, while others are beyond the limits of WHO[2] and ISSI[3] (Indian Standard Specification for drinking water).

Key Words: Physico-chemical parameters, ISSI (Indian Standard Specification for drinking water), APHA, T.D.S., B.O.D., C.O.D.

INTRODUCTION

Water bodies of the earth are being continuously polluted by a variety of sources. The pollution is occurring in all types of water bodies; both freshwater bodies like ponds, lakes and rivers as well as marine bodies like coastal and deep-water seas. Pollution is defined as "the introduction by man into the environment of substance or energy liable to cause hazards to human health, harm to living resources and ecological damage, or interference with legimate uses of the environment".[4]

Pollution: Development of economically acceptable, efficient and appropriate pollution control technologies should be given priority.[5] Major causes of encroachment of waterbodies are deposition of acid, organic chemical, inorganic minerals, pesticides, plant nutrient, sewage, detergents, agricultural chemicals, industrial effluents, silt, oil, radioactive substances and heat into the water bodies. Increased organic matter can alter energy relationships in the stream, disrupting biotic community structure and function.[6]

Assessing the effects of effluent discharge on the health of receiving waterways is of considerable environmental consequence, especially in catchments with variable stream flows and where population pressure through urbanization and periurbanization is placing increasing pressure upon WWTP infrastructure and the health of freshwater ecosystems.[7] Increased nutrient loading can lead to eutrophication[8] and temporary oxygen deficits.[9] Deposition of acidic substances causes acidification of water by lowering its pH below 6.0. The sulphates, nitrates and chlorides have been reported to make water bodies like lakes, rivers, dams and ponds acidic in many countries. Hardness is an important basis for the classification of water.[10] It indicates the suitability of water for the purposes used. Hardness is caused by Calcium and Magnesium ions in water.

The objective of the present study is to determine various physico – chemical parameters of Kanke dam water, as has not yet been explored scientifically. Therefore requires monitoring of various water quality parameters i.e. temperature, pH, turbidity, total dissolved solids (T.D.S.), total alkalinity, hardness, calcium, magnesium,

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sulphate, phosphate, chloride, nitrate, B.O.D. (Biological Oxygen Demand) and C.O.D. (Chemical Oxygen Demand).

MATERIALS AND METHODS

Kanke Dam: The study area Kanke Dam is one of the important water reservoir of Ranchi through which drinking water is supplied to various parts of Ranchi. It was constructed by DVC (1954- 1956). Drainage area of the dam is 9.44sq. miles. Capacity of reservoir is 7.05 ac feet. Estimated daily supply through the dam is 47 lac gallons. Kanke dam has maximum height of 53' and length is 4700'. Water of Kanke dam is widely used by a large number of people of Ranchi for drinking, bathing, washing fishing etc. The immunological investigation of Kanke Dam was started from January 2013. On the basis of preliminary observation and for the convenience, 4 experimental stations or sites were designated as site A, site B, site C and site D. These sites were chosen on the basis of the magnitude of activities and disturbances.

Biochemical Analysis of effluent: Samples for analysis with standard procedure in accordance with standard method.[1,11] The instruments has used in the limit of precise accuracy and chemicals used of G.R. Grade. Sampling was done at regular intervals of every month. At sampling site various physical parameters like appearance, colour, odour, pH and temperature has been observed and noted down on the spot immediately after the collection. Air and water temperature by mercury thermometer, pH by digital pH meter and total dissolved solid (T.D.S.) by pen type TDS meter. For chemical variables of water like total alkalinity and total hardness etc, the samples were collected in B.O.D. bottle, stored at 4 C and analysed in laboratory. The T-H, Ca-H, Mg-H has measured titrimetrically by using EDTA, Chloride by Mohr's Argentrometric titration and K_2CrO_4 as indicator, Total alkalinity has determined by titrimetric methods using phenophthalein and methyl orange indicators. Sulphate, phosphate, Chloride and Nitrate measured by spectrophotometer.

OBSERVATIONS

PARAMETERS	UNITS	Desirable Limit	SAMPLES		
			MAX	MIN	AVG
Temperature	0^0C	Shall not exceed 50^0 C above the receiving water temperature	40.30	7.20	12.30
pH	--	6.5-8.5	7.10	8.30	7.71
Turbidity	NTU	5-10	78.70	13.70	43.92
Total Solid	ppm	500-600	1260	260	585.34
Total Alkalinity	ppm	200-600	89	248	170.38
Hardness	ppm	300-600	31.90	91.60	54.45
Ca hardness	ppm	75	43.40	97.80	74.86
Mg hardness	ppm	30	6.50	26.10	16.35
Sulphate	ppm	150	8.70	33.90	20.79
Phosphate	ppm	5	0.08	0.61	0.30
Chloride	ppm	250	11.80	44.10	30.09
Nitrate	ppm	50	0.217	1.325	0.619
B.O.D	ppm	30	2	5.92	4.31
C.O.D	ppm	250	3	46	17

RESULTS

The present study has provided information about the variation in water characteristics of Kanke dam. General observations showed that water samples of Kanke dam are muddy in appearance, brownish-green in colour, salty in taste and its odour is pungent. All values of studied parameters are present as follows pH 7.71, temperature 12.30 C, turbidity 45.70 NTU, T.D.S. 585.34 ppm, total alkalinity 170.38 mg/l, hardness 54.45, calcium 74.86 mg/l, magnesium 16.35 mg/l, sulphate 8.70 mg/l, phosphate

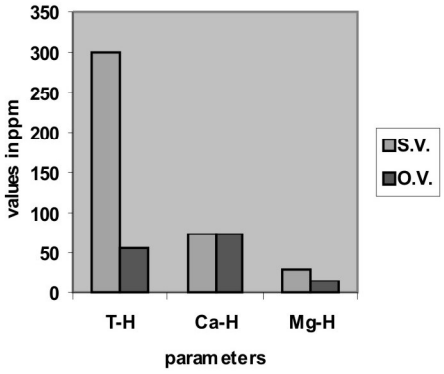


Fig 1: Histogram showing deviations in T-H, Ca-H and Mg-H in Kanke Dam

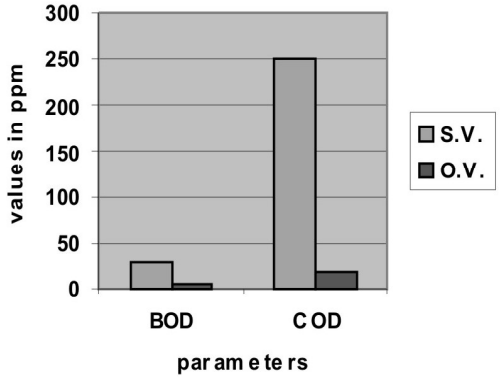


Fig 2: Histogram showing deviations in B.O.D. and C.O.D. Kanke Dam water.

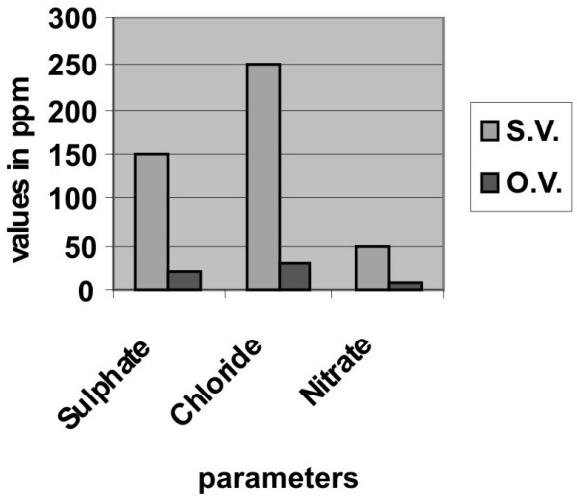


Fig 3: Histogram showing deviations in Sulphate, Chloride and Nitrate in Kanke Dam water.



Fig 4: Showing the accumulation of waste around Kanke Dam due to anthropogenic activities.
S.V.= Standard Values, O.V.=Observed Values

0.08 mg/l, chloride 30.09 mg/l, nitrate 0.619 mg/l, BOD 4.31 mg/l and COD 17.98 mg/l.

DISCUSSION

In the present investigation of Kanke dam water, most of the values of some physico-chemical parameters exceed the desirable limit according to ISSI (Indian Standard Specification for drinking water). pH of water is influenced by the presence of total alkalinity and carbon-di ioxide. Alkaline nature of inland water has been reported.[12] Increases in hardness of water is due to the natural accumulation of salts mainly calcium and magnesium. High concentration of salts and chlorides also get added to the water from the discharge of industrial effluents. Calcium is always present in natural water in the form of carbonate. Human activities and addition of nutrients increase the value Calcium. [13] The low values of Sulphate, Phosphate, Chloride and Nitrate in the water of Kanke dam are mainly due to discharge of industrial effluents or contamination with sewage water. The increased alkalinity was due to high concentration of nutrients in water of dam with run-off rainwater from the polluted soil. Rainfall is responsible for increasing the nitrates in water. [14]

More growth of micro organisms, plants and animals depletes oxygen. There is entry of sewage rich in bacteria.

So the bacteria utilized the dissolved oxygen by the process of biodegradation. In oligotropic water bodies, the amount of dissolved nutrient, salts remains low, therefore it supports spare plant and animal lives. This result in high dissolved oxygen gradually increased with depth. Besides in eutrophic water bodies, the organic matter accumulate abundantly which in turn are subjected to microbial decomposition. Therefore in Kanke dam water lower values of B.O.D. and C.O.D. indicate the eutrophication of Kanke Dam due to anthropogenic activities. The factors on which BOD depends are concentration of organic matter, temperature, plankton population, density and population of micro-organisms. [15] The analysis of sample water of Kanke dam was done, following the standard methods recommended.[16]

In the present study some of the parameters are found within the prescribed limits of IS : 10500 e.g. pH, temperature, T.D.S., Ca-H While most are beyond the limits alkalinity, T-H, Mg-H, Sulphate, Phosphate, Chloride, Nitrate, B.O.D., C.O.D.

CONCLUSION

This result indicated that Kanke dam water is non-portable for drinking. Permanent conservation strategies through scientific tools must be adopted for checking the entry of waste water into the Kanke dam and thereby keeping the water healthy and hygienic.

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